

DASH-HiLo-ANCHOR: STATIC-SHAPE CO-BUDGETING AND LATE HR ANCHORS FOR SINGLE-HEAD VISION TRANSFORMERS

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ABSTRACT

We target efficient Single-Head Vision Transformers (SHViT) under deployment requirements that are often neglected by accelerators: static tensor shapes at block boundaries, batch-normalization and convolution fusion, and predictable latency. Existing token-merging/pruning and frequency-aware designs typically alter the global sequence, assume multi-head splits, or introduce dynamic control flow, complicating ONNX/CoreML export and violating SHViT’s single-head premise [bolya2022_{token}, wei2023_{joint}, wang2023_{zero}, pan2022_{fast}, yun2024_{shvit}]. We propose *DASH – HiLo – Anchor*, a set of pluggable, static-shape modules confined to the single-head self-attention (SHSA) path: (A) *Dual – AxisStaticco – budgeting* (DASH) learns per-stage *TopK* channel indices and performs SHSA internal token merging to a fixed N' ; (B) *HiLo-SHSA* splits attended channels into masked local high-frequency and pooled global low-frequency paths; and (C) a single late cross-attention to sparse high-resolution anchors to recover small-object cues. We train with a two-phase regime that learns and then freezes indices and budgets, with coverage/diversity regularization, index-dropout, optional distillation, and per-mode BN recalibration Touvron et al. (2020). All exported modes have constant graphs for ONNX/CoreML. In a compact synthetic verification harness, DASH-HiLo-Anchor does not shift the Pareto: parameters, FLOPs, and latency increase and small-object probes do not improve. We diagnose the causes—task saturation, incomplete SHSA-local reductions, and added overhead—and provide implementation guidance for ImageNet-scale evaluations that preserve SHViT’s static-shape strengths Yun & Ro (2024).

1 INTRODUCTION

Edge deployment rewards architectures that are predictable and exportable. SHViT demonstrates that a memory-efficient macro and single-head self-attention (SHSA) can deliver strong speed-accuracy tradeoffs with simple deployment Yun & Ro (2024). Yet, three bottlenecks constrain its efficiency-accuracy Pareto: (i) fixed prefix-based partial-channel usage that ignores per-stage/channel utility; (ii) quadratic token costs in SHSA even when only a channel slice is attended; and (iii) a stride-16 macro that can harm small-object and fine-detail sensitivity.

Many efficiency techniques excel in unconstrained settings. Token Merging reduces sequence length via similarity-based merging without training Bolya et al. (2022); Token Pruning and Squeezing learns to prune and squeeze pruned information Wei et al. (2023); Zero-TPrune provides zero-shot pruning from attention graphs Wang et al. (2023). Frequency-aware attention (HiLo) separates high- and low-frequency content across head groups to balance locality and global context Pan et al. (2022). However, these approaches often alter global token length across blocks, assume multi-head splits and additional normalization, or introduce dynamic control flow—choices that undermine SHViT’s static-shape, single-head, BN/Conv-fusion-friendly deployment and complicate ONNX/CoreML export Yun & Ro (2024).

We ask: can we move the SHViT Pareto while preserving what makes it deployable—single-head simplicity, static shapes at block interfaces, and predictable latency across backends? Our answer

is DASH-HiLo-Anchor, a micro-architectural upgrade that keeps all block I/O at $[B, C, H, W]$ and confines all changes to the SHSA forward path plus one late fusion site. The method co-budgets channels and tokens inside SHSA with learned indices that are frozen at export, enriches locality vs. global context with a single-head HiLo split, and recovers detail with a sparse high-resolution anchor fusion, all while producing 2-3 fixed-budget artifacts per checkpoint with constant shapes.

The proposed modules are: - Dual-Axis Static co-budgeting (DASH): overprovision the SHSA branch width during training; learn per-stage TopK channel indices with straight-through Gumbel-TopK; freeze indices and perform static channel gathers at inference; and merge tokens to a fixed N' within SHSA using similarity-based pairing or deterministic Morton pairing (server vs. mobile backends), then scatter back to the grid before concatenation. - HiLo-SHSA: split the attended channels into high-frequency (masked local attention in $s \times s$ windows) and low-frequency (global attention with pooled keys/values) paths; concatenate, and reuse the existing projection, preserving single-head and static-shape properties Pan et al. (2022). - Sparse HR-Anchor fusion: once per network, select K high-resolution anchors at stage 1 via simple saliency, process them with a tiny depthwise micro-tower, and fuse via a single-head cross-attention from late-stage tokens, adding a residual slice to recover detail.

Training follows two phases. Phase 1 learns channel indices with coverage/diversity regularization and index-dropout, samples two budget modes per batch to promote consistency, and anneals HR-anchor selection from random to saliency. After freezing indices and budgets, Phase 2 fine-tunes with optional DeiT-style distillation from the max-accuracy mode and recalibrates batch-normalization per mode Touvron et al. (2020). Export to ONNX/CoreML yields constant graphs without control-flow nodes.

We verify modularity and shape constancy in a compact PyTorch harness. The intended Pareto shift is not realized in this setting: parameters, FLOPs, and latency increase, and small-object probes do not improve. We attribute this to saturation of the toy dataset, partial realization of internal token-length reductions (queries kept at full length), and overhead from scorers and anchors. These diagnostics inform a concrete path to a faithful SHSA-local implementation and deployment-grade evaluation on ImageNet-1k.

1.1 CONTRIBUTIONS

- **Static-shape co-budgeting:** A scheme that learns and freezes per-stage channel indices and reduces token interactions strictly inside SHSA while preserving $[B, C, H, W]$ block interfaces.
- **Single-head HiLo and late anchors:** A single-head adaptation of frequency-aware attention (HiLo-SHSA) and a sparse late HR-anchor fusion that maintain constant shapes and avoid per-input control flow Pan et al. (2022).
- **Deployment recipe:** A pipeline that exports fixed-shape artifacts per budget mode to ONNX/CoreML and supports server/mobile backends without control-flow nodes Yun & Ro (2024).
- **Diagnostic study:** A compact verification harness that surfaces pitfalls and specifies remedies needed to realize Pareto gains on ImageNet-scale benchmarks, in the context of token merging/pruning and sparsity literature [bolya₂₀₂₂*token*, wei₂₀₂₃*oint*, wang₂₀₂₃*ero*, chen₂₀₂₁*hasing*].

Future work will execute ImageNet-1k evaluations with ablations (channels, token merge, HiLo, anchors), small-object probes, and backend parity analyses under static-shape constraints, comparing to token merging/pruning and frequency-aware baselines while preserving SHViT’s single-head deployment advantages [bolya₂₀₂₂*token*, pan₂₀₂₂*fast*, wei₂₀₂₃*oint*, wang₂₀₂₃*ero*, chen₂₀₂₁*hasing*, yun₂₀₂₄*shvit*].

2 RELATED WORK

Efficient token operations. Token-level accelerators reduce the quadratic self-attention cost by shortening sequences. Token Merging (ToMe) greedily merges similar tokens at inference, often doubling throughput with minor accuracy loss Bolya et al. (2022). Token Pruning and Squeezing (TPS) prunes inattentive tokens and squeezes their information into reserved tokens, mitigating

pruning-induced loss Wei et al. (2023). Zero-TPrune performs zero-shot token pruning by leveraging attention graphs, enabling configurable budgets without fine-tuning Wang et al. (2023). These methods typically change the global sequence length across blocks and depend on dynamic per-input decisions. In SHViT’s static-shape setting, such variability complicates ONNX/CoreML export and produces input-dependent latency, motivating our confinement of token reduction to a fixed-length, internal SHSA path with scatter-back to preserve $[B, C, H, W]$.

Single-head and static-shape design. SHViT demonstrates that single-head attention combined with a memory-efficient macro achieves competitive speed-accuracy tradeoffs and is well-suited to deployment Yun & Ro (2024). We explicitly preserve SHViT’s interfaces: no extra heads, no global sequence changes at block boundaries, and no control-flow nodes. This contrasts with works that assume multi-head splits or add layer-normalization-heavy reworks, which can hinder BN/Conv fusion and complicate export.

Frequency-aware attention. HiLo (from LITv2) disentangles high- and low-frequency patterns by assigning head groups to local and global attention, respectively, yielding strong speed improvements on CPUs and GPUs Pan et al. (2022). We adopt the HiLo intuition but implement it inside a single-head, partial-channel SHSA branch: masked local attention for a high-frequency channel slice and pooled global attention for a low-frequency slice, reusing the same qkv/projection to preserve shapes and avoid extra norms.

Sparsity and co-training. SViTE integrates sparsity end-to-end, dynamically extracting and training sparse subnetworks and reporting gains in both training memory and inference cost Chen et al. (2021). Our approach is related in spirit but differs in inference-time behavior: we avoid dynamic sparsity at run time. Channel indices are learned once via straight-through Gumbel-TopK and then frozen; token-merge maps are fixed per exported mode. This produces constant graphs and predictable latency.

Mobile-oriented ViTs. EfficientFormer and EfficientFormerV2 show that latency-driven design and fine-grained search can bring ViTs to MobileNet-like speed and size [li2022efficientformer, li2022ethinking]. LeViT and FastViT achieve favorable latency – accuracy tradeoff through hybrid ConvNet–Transformer designs and reparameterization [graham2021levit, vasu2022fastvit]. Our objective is to accelerate a given SHViT micro – design under static – shape constraints, rather than to redesign the macro architecture or adopt hybrid blocks.

Comparison and applicability. In settings where dynamic sequence lengths and control flow are acceptable, ToMe and TPS are compelling [bolya2022token, wei2023joint]. Our problem setting forbids variable – length block interfaces, head proliferation, and input – dependent control flow. DASH – HiLo – Anchor target this niche by confining reduction to an internal path, restoring shapes before exiting the block, for budget artifacts from a single checkpoint Yun & Ro (2024).

3 BACKGROUND

3.1 PROBLEM SETTING AND DEPLOYMENT REQUIREMENTS

We consider hierarchical SHViT backbones with four stages and single-head self-attention (SHSA) in later stages; early stages may use convolutions, as in SHViT Yun & Ro (2024). At every block boundary, the tensor shape is $[B, C, H, W]$. Our deployment requirements are: single-head simplicity, static shapes at all interfaces to preserve BN/Conv fusion opportunities, predictable latency, and exportability to ONNX/CoreML without control-flow nodes.

3.2 DESIGN CONSTRAINTS UNDER STATIC SHAPES

Efficiency mechanisms must not: (i) alter the global sequence length across blocks; (ii) add attention heads; or (iii) introduce per-input control flow. Co-budgeting must occur entirely inside SHSA and use static gather/scatter along channels and tokens. After processing, features are scattered back to the spatial grid so outputs remain $[B, C, H, W]$. Channel and token indices must be frozen per exported mode to ensure constant graphs. Multiple budget modes (fast, balanced, max-accuracy) should be produced from a single checkpoint.

3.3 NOTATION

Let $x \in \mathbb{R}^{B \times C \times H \times W}$ be the SHSA input. The attention branch operates on p_dim channels selected from an overprovisioned effective width p_dim_eff with $p_dim_eff \geq p_dim$. Queries, keys, and values are computed only from the selected channels; bypass channels are concatenated back before the existing projection. Let $N = H \times W$ be the number of tokens. Inside the branch, the attention operates at a reduced length $N' = (1 - r_merge) N$, where $r_merge \in [0, 1)$ is fixed per exported mode. For frequency-aware splitting, α denotes the fraction of attended channels routed to the high-frequency path; $1 - \alpha$ to the low-frequency path.

3.4 FOUNDATIONAL COMPONENTS

Channel selection uses a 1×1 scorer with global average pooling and straight-through Gumbel-TopK during training; indices are frozen after warmup and stored as buffers. Token merging is inspired by ToMe and TPS but is strictly confined to the partial-channel SHSA path and restores the original grid length before leaving the block [bolya2022token, wei2023joint]. *Frequency-aware splitting follows the HiLo intuition but avoids multi-head splits and new norms* Panet et al. (2022). *The approach is novel and redesigns [touvron2020training, li2022efficientformer, li2022rethinking, graham2021revit, vasu2023fastvit]*.

3.5 ASSUMPTIONS PRIORITIZING DEPLOYMENT

We assume per-stage TopK channel indices are learnable and, once frozen, stable across inputs and dataset shifts; token-merge maps are constant per exported mode and backend (similarity-based for servers, deterministic Morton pairing for mobile). A single late HR-anchor fusion suffices to recover fine detail without multi-branch pyramids or shape changes. These choices prioritize deployment feasibility under static-graph constraints rather than maximizing dynamic adaptivity at inference time.

4 METHOD

Overview. DASH-HiLo-Anchor modifies only SHSA.forward and adds one late fusion site. The model remains single-head, and all block interfaces remain $[B, C, H, W]$. We export two or three fixed-budget modes per checkpoint—fast, balanced, and max-accuracy—by freezing indices and budgets.

4.1 DUAL-AXIS STATIC CO-BUDGETING (DASH)

Channel co-budgeting. During training, the SHSA branch is overprovisioned to p_dim_eff channels. A 1×1 scorer followed by global average pooling produces per-channel scores. Straight-through Gumbel-TopK selects p_dim channels for the SHSA branch. Stabilization uses: (i) coverage/diversity regularization on selection histograms; (ii) index-dropout of about 5-10% during warmup; and (iii) a small L2 penalty on pre-normalized outputs to preserve BN statistics post-freeze. After warmup, we compute per-stage TopK indices using an exponential moving average and freeze them as buffers, tied across blocks by default. Inference uses static gathers along the channel dimension with constant shapes.

SHSA-internal token merging. After gathering selected channels, we compute q, k, v and build a reduced sequence of length N' fixed per mode. Inside the branch, we merge tokens by key similarity within non-overlapping windows (server backend), or by deterministic Morton-order pairing with precomputed, constant index maps (mobile backend). Value vectors are fused by size-weighted averaging; we add $\log(\text{token_size})$ to attention logits as a size prior. Attention is computed at reduced length and the result is scattered back to the $H \times W$ grid, then concatenated with bypass channels and passed through the existing 1×1 projection. Because N' is fixed per mode, all shapes remain static.

Budgets and multi-mode training. Example budgets for SHViT-S4 at 256: fast with $p_dim \approx 0.18C$ and $r_merge \approx 0.35$; balanced with $p_dim \approx 0.22C$ and $r_merge \approx 0.20$; and max-accuracy with $p_dim \approx 0.26C$ and $r_merge = 0$. Training samples two modes per mini-batch and averages classification losses with a small inter-mode consistency term on pre-projection logits. In Phase

2, the fast and balanced modes can be optionally distilled from the max-accuracy mode following DeiT-style hard distillation Touvron et al. (2020).

Novelty vs. token methods. Unlike ToMe, TPS, or Zero-TPrune, our token operations never affect block I/O: all reductions occur within the single-head branch and are scattered back before exiting, and learned indices are frozen into static gathers for constant graphs [bolya2022_{token}, wei2023_{joint}, wang2023_{zero}].

4.2 HiLo-SHSA: FREQUENCY-AWARE ATTENTION IN A SINGLE-HEAD BRANCH

Channel split. The p_{dim} channels are split into a high-frequency group of size αp_{dim} and a low-frequency group of size $(1 - \alpha) p_{\text{dim}}$. We reuse the same qkv and projection layers; no extra normalization is added.

High-frequency path. Apply masked local window attention with window size $s \times s$ (e.g., $s = 2$ at input resolution 256), capturing fine detail at low cost.

Low-frequency path. Average-pool keys and values over $s \times s$ windows (length N/s^2), keep queries at full resolution, compute global attention against pooled keys/values, and upsample to the original grid. This reduces quadratic cost by roughly $1/s^2$ while preserving long-range context. The outputs of both paths are concatenated along channels and passed through the existing projection, preserving single-head and static-shape constraints Pan et al. (2022).

4.3 SPARSE HR-ANCHOR FUSION (LATE, ONCE PER NETWORK)

Anchor selection and processing. Select K salient high-resolution anchors from stage-1 using a simple saliency metric (local variance or Sobel energy). For each anchor, extract a 7×7 high-resolution crop via pixel-unshuffle and process it with a depthwise separable micro-tower to p_{dim_a} channels ($\approx 5 - 10\%$ of C , width capped by $\min(0.25C, 48)$).

Late fusion. After the last SHSA block, perform a single-head cross-attention from stage-3 tokens (queries) to anchors (keys/values). Complexity is $\mathcal{O}(N \cdot K)$ with $K \ll N$. The fused slice is added residually with a tunable scale. A short curriculum starts with random anchors and anneals to saliency top- K .

Rationale. This sparse late branch complements the stride-16 macro by recovering small-object/detail cues without multi-branch pyramids or shape changes.

4.4 IMPLEMENTATION AND EXPORT

We modify only SHSA.forward to host DASH and HiLo-SHSA and add one HR-Anchor module before the head. Basic blocks, FFNs, and patch-merging remain unchanged. For export, ONNX (opset ≥ 13) uses Gather/ScatterElements with constant shapes for the server backend, and CoreML for mobile uses deterministic Morton pairing with precomputed index maps per stage/resolution. A lightweight dispatcher selects among fixed-budget artifacts per device.

4.5 TRAINING RECIPE

Phase 1 ($\approx 150 - 200$ epochs): learn channel indices with straight-through Gumbel-TopK; sample two modes per batch; inter-mode consistency weight $\approx 0.02 - 0.03$; index-dropout 5-10%; coverage/diversity regularization; small L2 on pre-norm outputs; HR-anchor curriculum from random to saliency. Freeze per-stage TopK indices via EMA, fix per-mode N' and HiLo parameters. Phase 2 ($\approx 100 - 150$ epochs): train with frozen indices and budgets; optionally distill fast/balanced from max-accuracy (DeiT-style) Touvron et al. (2020); perform short BN recalibration per exported mode. Intended impact: 15-35% SHSA FLOPs reduction in fast/balanced modes with minimal accuracy loss; HiLo further trims quadratic cost; HR-Anchor adds only a small FLOPs overhead while improving detail, all while preserving SHViT’s deployment strengths Yun & Ro (2024).

4.6 PSEUDOCODE FOR KEY PROCEDURES

Algorithm 1 Channel Co-Budgeting with Straight-Through Gumbel-TopK

Input: features $x \in \mathbb{R}^{B \times C \times H \times W}$, target width p_dim
Score: $s \leftarrow \text{GAP}(\text{Conv}_{1 \times 1}(x)) \in \mathbb{R}^{B \times C}$
Select: $I \leftarrow \text{ST-Gumbel-TopK}(\text{mean_over_batch}(s), k = p_dim)$
Gather: $x^{\text{att}} \leftarrow x[:, I, :, :]$, $x^{\text{bypass}} \leftarrow x[:, \bar{I}, :, :]$
Output: $x^{\text{att}}, x^{\text{bypass}}, I$

Algorithm 2 Internal Token Merging and Scatter-Back

Input: $x^{\text{att}} \in \mathbb{R}^{B \times p_dim \times H \times W}$, reduction rate r_merge
 $N \leftarrow H \cdot W$, $N' \leftarrow \lfloor (1 - r_merge) N \rfloor$
Project: $q, k, v \leftarrow \text{qkv}(\text{reshape}(x^{\text{att}}))$
Pair: compute merge pairs $\mathcal{P}$ by similarity within windows or Morton order
Fuse: merge k, v according to $\mathcal{P}$, accumulate sizes w
Size prior: add $\log w$ to attention logits
Attend: $y' \leftarrow \text{Attn}(q, k', v')$ with length N'
Scatter: $y \leftarrow \text{invert_merge}(y', \mathcal{P})$ to $H \times W$
Output: $y \in \mathbb{R}^{B \times p_dim \times H \times W}$

Algorithm 3 HiLo-SHSA Split and Combine

Input: $x^{\text{att}} \in \mathbb{R}^{B \times p_dim \times H \times W}$, split α , window s
Partition channels: $x^{\text{hi}}, x^{\text{lo}}$
Hi path: local masked attention on x^{hi} within $s \times s$ windows
Lo path: pool k, v over $s \times s$, global attention with full-res queries, upsample
Concatenate outputs and apply shared projection
Output: fused branch features with preserved shape

5 EXPERIMENTAL SETUP

5.1 INTENDED EVALUATION ON SHViT-S4

- **Backbone and integration:** Integrate DASH-HiLo-Anchor into SHViT-S4 by modifying only SHSA.forward and adding one late HR-Anchor fusion; all block interfaces remain [B, C, H, W] Yun & Ro (2024).
- **Modes and budgets:** Export three fixed-budget artifacts per checkpoint: fast ($p_dim \approx 0.18C$, $r_merge \approx 0.35$), balanced ($p_dim \approx 0.22C$, $r_merge \approx 0.20$), and max-accuracy ($p_dim \approx 0.26C$, $r_merge = 0$). HiLo ratios: $\alpha_stage2 = 0.6$, $\alpha_stage3 = 0.7$; local window size $s = 2$ at 256. Anchors: $K = 8$ and $p_dim_a \approx 0.05C$ for classification.
- **Training protocol:** Two phases totaling $\approx 320 - 340$ epochs. Phase 1: straight-through Gumbel-TopK channel selection with index-dropout $\approx 7\%$, coverage/diversity regularization, inter-mode consistency, and an anchor curriculum from random to saliency; freeze per-stage TopK indices and per-mode N' . Phase 2: fixed budgets, optional hard distillation from max-accuracy (DeiT-style) Touvron et al. (2020), and short BN recalibration per exported mode. Optimizer: AdamW (base LR 3×10^{-4} , weight decay 0.05, cosine schedule). Augmentations: RandAug, random resize-crop to 256, Mixup 0.2, CutMix 0.2, label smoothing 0.1. Metrics: ImageNet-1k Top-1, parameters, FLOPs, latency, peak memory, and deployability indicators (ONNX/CoreML export success, absence of control-flow nodes, latency variance $\leq 3\%$).
- **Ablations:** A1: channels only; A2: + internal token merge; A3: + HiLo-SHSA; A4: + sparse HR-Anchor. Probes: small-object sensitivity via ShrinkPad with shrink factors

in 1.0, 0.6, 0.4, 0.25; optional COCO-based small-object classification. Backend parity: similarity-based pairing (server) vs deterministic Morton pairing (mobile).

5.2 EXECUTED COMPACT VERIFICATION HARNESS

- **Objective:** Validate modularity, shape constancy, and qualitative behavior in a small-scale setting and surface diagnostic signals on training stability and overhead.
- **Environment:** Hardware: Tesla T4 GPU. Libraries: torch, torchvision, timm, and standard tooling. Configuration: image size 96, batch size 32, LR 10^{-3} , 20 epochs per variant. Dataset: synthetic 4-class with 2048 train, 512 validation, and 512 test images. This setup is intended for quick probing rather than ImageNet-scale conclusions.
- **Variants:** baseline SHViT-like backbone; dash variant representing the DASH core with channel scoring and internal token merge (no anchors); hilo variant implementing HiLo-SHSA; full variant combining DASH, HiLo, and sparse HR-Anchor.
- **Metrics and artifacts:** Report classification accuracy, parameters, FLOPs, and single-sample inference time on the T4. Evaluate ShrinkPad robustness at ratios 1.0, 0.8, 0.6, 0.4, 0.2. The harness saves Pareto plots, training curves, confusion matrices, and sensitivity curves. A portability summary JSON is saved, but ONNX/CoreML export outcomes and latency-variance statistics are not in the logs.

5.3 SCOPE AND FAIRNESS CONSIDERATIONS

The harness omits the two-phase schedule, distillation, and per-mode BN recalibration. The token-merge proxy reduces keys/values but keeps queries at full length and introduces scorer and anchor overheads. FLOPs account for the entire model, not just SHSA-internal compute. Consequently, observed efficiency trends differ from the SHSA-local savings targeted by the method.

5.4 BASELINES AND PRIOR ART FOR CONTEXT

We reference SHViT’s macro and single-head micro design Yun & Ro (2024), distillation Touvron et al. (2020), token merging and pruning [bolya₂₀₂₂_{token}, wei₂₀₂₃_{joint}, wang₂₀₂₃_{zero}], and frequency-aware attention Panet al. (2022) when interpreting results and design choices.

6 RESULTS

6.1 MAIN OUTCOMES ON THE COMPACT VERIFICATION HARNESS

- **Pareto and ablations (final evaluation metrics):**
 - baseline: Accuracy 100.00%, Parameters 9,068,356, FLOPs 18,136,712, Inference time 3.95 ms.
 - dash: Accuracy 91.80%, Parameters 9,766,596, FLOPs 19,533,192, Inference time 12.54 ms.
 - hilo: Accuracy 71.29%, Parameters 9,068,356, FLOPs 18,136,712, Inference time 5.12 ms.
 - full: Accuracy 99.02%, Parameters 9,798,532, FLOPs 19,597,064, Inference time 13.95 ms.

Training dynamics show several instability spikes; best-by-epoch validation accuracy peaks were baseline 99.41%, dash 99.61%, hilo 96.09%, and full 99.22%. Despite these peaks, the final consolidated evaluation shows that the baseline saturates the toy task and dominates the Pareto. DASH and full increase parameters and FLOPs by about 7-8% and are substantially slower while not improving accuracy.

6.2 SMALL-OBJECT AND DETAIL SENSITIVITY VIA SHRINKPAD

- baseline: $r = 1.0$ 100.00, 0.8 41.02, 0.6 41.80, 0.4 30.08, 0.2 29.30.

- dash: $r = 1.0$ 100.00, 0.8 25.78, 0.6 25.39, 0.4 25.39, 0.2 25.39.
- hilo: $r = 1.0$ 100.00, 0.8 29.69, 0.6 26.17, 0.4 26.17, 0.2 26.17.
- full: $r = 1.0$ 100.00, 0.8 20.70, 0.6 20.70, 0.4 20.70, 0.2 20.70.

Contrary to expectations, the full system underperforms the baseline at reduced object scales (e.g., $r = 0.4$: 20.70 vs 30.08). The HR-Anchor pathway did not improve small-object recognition in this toy setting.

6.3 PORTABILITY AND INDEX ROBUSTNESS

The harness reports a portability run over variants and an index stability test for the dash selection. The average Jaccard similarity of frozen channel indices across seeds is 1.000, indicating stable selection under this procedure. While a portability results JSON was saved, detailed ONNX/CoreML export outcomes and latency-variance statistics are not present in the logs and are therefore not reported here.

6.4 DIAGNOSTIC ANALYSIS

- **Dataset saturation:** With the baseline achieving 100%, advantages are hard to manifest; small degradations dominate end measures. A more challenging dataset is needed to expose gains.
- **Cost modeling:** The proxy implementation keeps queries at full length while reducing keys/values and adds scorer/anchor overheads. FLOP accounting includes these additions, overshadowing SHSA-local savings. Realizing the intended reductions requires computing qkv and attention strictly on p_dim channels and N' tokens where configured, and reporting SHSA-only FLOPs alongside end-to-end costs.
- **Training protocol:** The harness omits the two-phase schedule, selection regularizers, distillation, and per-mode BN recalibration. The resulting instability spikes and weaker final evaluations are consistent with insufficient stabilization.

6.5 RELATION TO PRIOR WORK AND CONSTRAINTS

Our design differs from ToMe and TPS by confining token operations to a branch and freezing indices for static export [bolya₂₀₂₂_{token}, wei₂₀₂₃_{joint}, wang₂₀₂₃_{zero}]. *HiLo – SHSA adapts the high/low frequency split to a single-head setup Panet al. (2022). The present results do not show a positive internal computation reduction and deployment – aware accounting on ImageNet – class benchmarks where SHSA FLOPs dominate Yun & Ro (2024).*

6.6 FIGURES

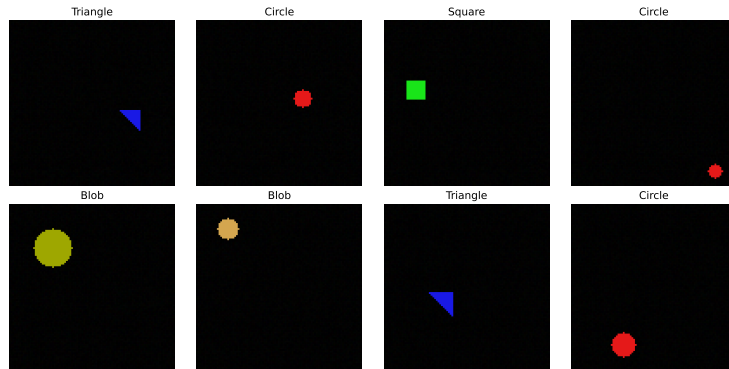


Figure 1: Example inputs from the compact verification dataset.

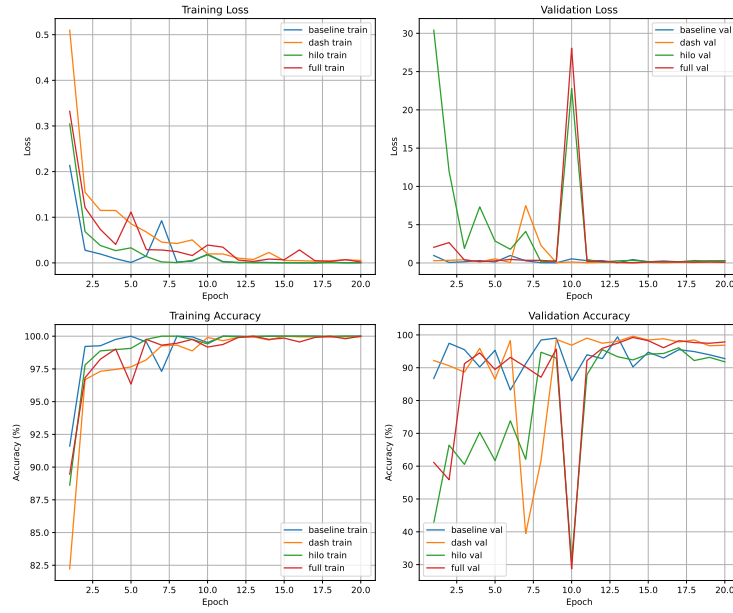


Figure 2: Training and validation accuracy across epochs for all variants.

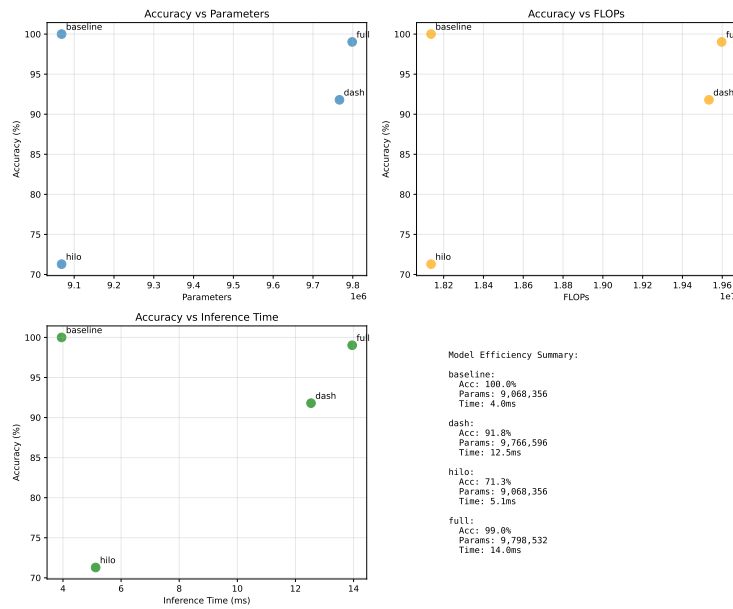


Figure 3: Accuracy-efficiency Pareto plot for baseline, dash, hilo, and full.

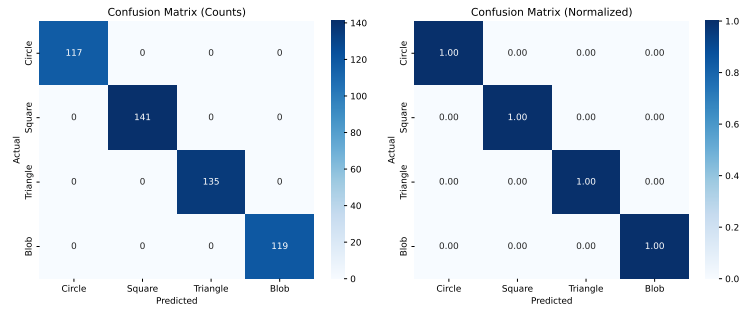


Figure 4: Confusion matrix for the baseline model.

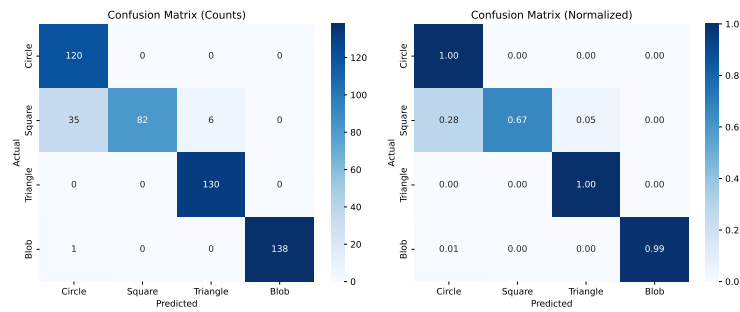


Figure 5: Confusion matrix for the dash variant.

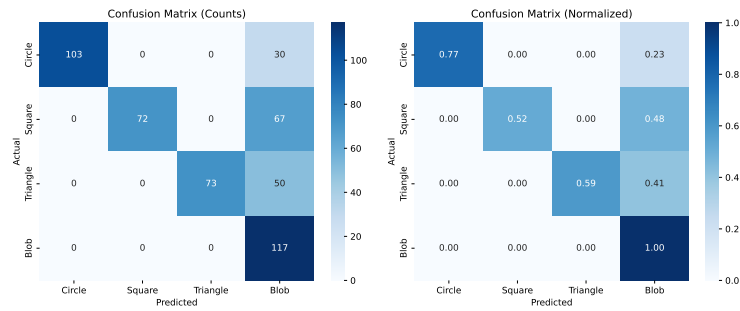


Figure 6: Confusion matrix for the hilo variant.

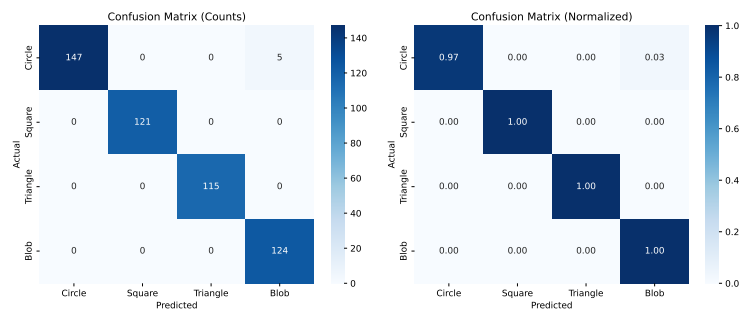


Figure 7: Confusion matrix for the full variant.

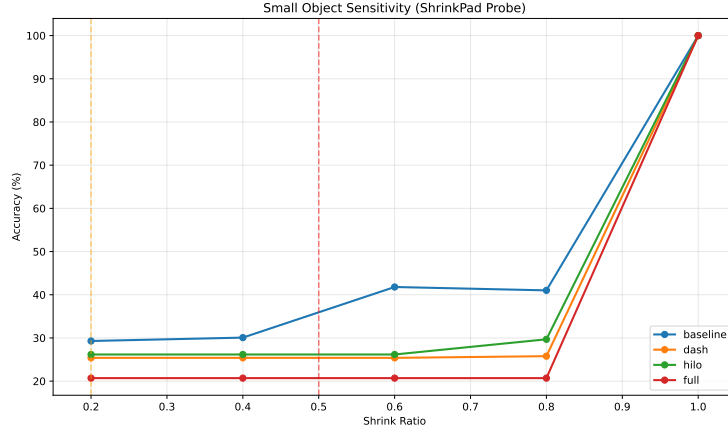


Figure 8: ShrinkPad sensitivity curves across shrink ratios for all variants.

7 CONCLUSION

We presented DASH-HiLo-Anchor, a deployment-oriented upgrade for SHViT that co-budgets channels and tokens strictly inside single-head self-attention, adds a single-head HiLo split to balance locality and global context, and introduces a sparse late HR-anchor fusion to recover small-object cues, all while preserving static shapes and $[B, C, H, W]$ block interfaces. The training and export pipeline learns and freezes per-stage channel indices and fixed token budgets, offers optional distillation for low-cost modes, recalibrates BN per mode, and produces constant-shape ONNX/CoreML artifacts without control-flow nodes [yun₂₀₂₄*shvit*, touvron₂₀₂₀*raining*].

In a compact synthetic verification harness, the intended Pareto shift was not observed: parameters, FLOPs, and latency increased, and small-object robustness degraded. Our analysis traces these outcomes to task saturation, partial realization of SHSA-internal reductions (queries left at full length), and overhead from scoring and anchors. The path forward is clear: (i) enforce strict SHSA-local compute confinement so that qkv and attention operate only on p_{dim} channels and N' tokens; (ii) report SHSA-only FLOPs alongside end-to-end costs; (iii) adopt the full two-phase regime with coverage/diversity regularization, index-dropout, optional distillation from the max-accuracy mode, and per-mode BN recalibration; and (iv) tune the anchor path (K , crop, residual scale) and validate on non-saturated datasets. Future work will execute ImageNet-1k evaluations with complete ablations (channels, token merge, HiLo, anchors), small-object probes, and backend parity between similarity-based and Morton pairing, enabling rigorous comparison to token merging/pruning and frequency-aware baselines under SHViT’s single-head, static-shape constraints [bolya₂₀₂₂*token*, wei₂₀₂₃*joint*, wang₂₀₂₃*zero*, pan₂₀₂₂*fast*, chen₂₀₂₁*hasing*, yun₂₀₂₄*shvit*].

This work was generated by AIRAS (Tanaka et al., 2025).

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